

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 7 Winter 2025 ( Regular )**

**Course :Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Technology**

**Semester : SEMESTER - 7**

**Subject Code & Name: BTETOE703D - MOBILE COMPUTING**

**Time : 3 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

**Q1. Objective type questions. (Compulsory Question)**

**12**

- 1 What is the primary difference between Mobile Computing and Wireless Networking?  
a) Mobility focus b) Same c) Wired only d) Better connectivity
- 2 What is the typical structure of a Mobile Computing Application?  
a) Presentation Application Data b) Client only c) Server only d) No layers
- 3 Which of the following is an example of Random Assignment Scheme?  
a) TDMA b) CDMA c) Slotted ALOHA d) FDMA
- 4 What is a common issue in Wireless MAC protocols?  
a) Hidden terminal b) Unlimited bandwidth  
c) No collisions d) Centralized always
- 5 What is a key feature of Mobile IP?  
a) Route Optimization b) Fixed IP c) No handover d) No tunneling
- 6 Which generation does GSM belong to?  
a) 2G digital b) 3G c) 1G analog d) 4G
- 7 What does GPRS primarily provide?  
a) Circuit voice b) Packet data c) No data d) Analog
- 8 Which are popular Routing Protocols in MANET?  
a) AODV DSR b) OSPF only c) BGP only d) Static
- 9 What is the main difference between VANET and MANET?  
a) No mobility b) Low speed c) High mobility d) Same
- 10 What are special constraints for Mobile Device Operating Systems?  
a) Limited resources b) Unlimited power c) High processing d) No touch
- 11 What is a major security issue in M-Commerce?  
a) Transaction privacy b) Unlimited security c) No threats d) Physical theft
- 12 What is a critical security issue in mobile payment systems?  
a) Data privacy b) No encryption  
c) Unlimited access d) Physical security only

- Q2. Solve the following.
- A) Describe the key characteristics of Mobile Computing. Explain how these characteristics impact the design of mobile applications. 6
  - B) Draw and explain the typical structure (architecture) of a Mobile Computing Application 6
- Q3. Solve the following.
- A) Explain the major issues in Wireless MAC protocols with suitable diagrams. 6
  - B) Compare Fixed Assignment Schemes, Random Assignment Schemes, and Reservation-Based Schemes for wireless MAC. Provide one example for each category. 6
- Q4. Solve Any Two of the following.
- A) Explain the features of Mobile IP and describe its key mechanisms with a neat diagram. 6
  - B) Describe the route optimization process in Mobile IP. Why is it necessary and how does it improve performance? 6
  - C) Compare the architecture and services provided by GSM, GPRS, and UMTS. 6
- Q5. Solve Any Two of the following.
- A) Explain the basic concepts and characteristics of Ad-Hoc Networks. 6
  - B) Describe the design issues and routing challenges in Mobile Ad-Hoc Networks (MANET). 6
  - C) Discuss the key differences between MANET and VANET. 6
- Q6. Solve Any Two of the following.
- A) Explain the special constraints and requirements that influence the design of Mobile Device Operating Systems. 6
  - B) Describe the structure of M-Commerce. Explain its advantages and disadvantages with real-world examples. 6
  - C) Discuss the key components of a Mobile Payment System. Explain the major security issues in M-Commerce and suggest possible solutions. 6

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